AARON MILLER

Full-Stack Engineer (AI-Forward) — Startup Generalist

📍 Seattle, Washington | 📧 aaaronmiller@gmail.com | 💼 linkedin.com/in/aaaronmiller | 🔗 github.com/aaaronmiller | 🌐 aaaronmiller.github.io/portfolio

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PROFESSIONAL SUMMARY

Full-Stack Engineer (AI-Forward) and owner/operator who ships production software in days, not months.

I cover the full surface area early-stage teams need one person to cover: **Svelte / SvelteKit / React / React Native + Tailwind** on the frontend, **Hono / Express / Node / Bun** on the backend, **Postgres / Redis / D1 (Cloudflare)** for data, and **JS / TS / Python** across the stack — with **API integration, production deployment, and edge deployment (Cloudflare Workers, 200+ edge locations)** as core muscles, not afterthoughts.

For **17+ years as owner/operator** (Aaron Miller Computer Solutions, 2008–Present) and **Lead Front End Developer at 2FIX.US (2014–Present)**, I've owned the entire **client lifecycle: concept → design approval → alpha → production → SEO → maintenance and content updates** — finding clients, scoping work, shipping, and keeping it running. That breadth is the startup job: scope ambiguous problems, prototype fast, ship to prod, monitor, and iterate with users.

I prototype **working, deployed features in days** and harden them for production — robust error handling, cost tracking, and 99.9% uptime patterns proven while processing **300M+ tokens, 500+ ops/hour, and 100K+ data points/day** across AI-forward systems. Recent contract work (2019–2023) adds classic full-stack evidence: **e-commerce serving 10K+ monthly users, real-time logistics dashboard (React/WebSocket), and Python/AWS Lambda pipelines handling 1M+ records/month.**

Target Roles: Senior Software Embedded Engineer • Axon

CORE STACK — AT A GLANCE

Layer	What I Ship With
Frontend	Svelte, SvelteKit, React, React Native, TailwindCSS, HTML5/CSS3/SASS, Handlebars, jQuery/Bootstrap (legacy breadth)
Backend	Hono, Express.js, Node.js, Bun, RESTful APIs, WebSocket, Cloudflare Workers (edge)
Data	PostgreSQL, Redis, D1 (Cloudflare), MySQL, MongoDB/Mongoose, Sequelize, Vector DBs (HNSW), Firebase
Languages	JavaScript, TypeScript, Python, SQL, Bash, PHP
Mobile	React Native
AI-Forward	Claude/GPT-4/Gemini/local LLMs, MCP, RAG, multi-agent orchestration, vector embeddings, cost/perf routing
Integration	Third-party API integration (Postman-driven), OAuth, MCP servers, Playwright browser automation
Deploy & Ops	Cloudflare Workers/D1, Vercel, AWS Lambda, Docker, GitHub Actions CI/CD, edge caching, Linux/macOS/Windows admin
Client Lifecycle	Concept diagrams (Balsamiq) → alpha → production → SEO → maintenance at 2FIX.US; CMS (WordPress/Drupal) when it ships faster

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PROFESSIONAL EXPERIENCE

Owner / Operator — Full-Stack Engineer

Aaron Miller Computer Solutions | Seattle, WA | May 2008 – Present (17+ years)

Self-directed practice owning every stage of delivery for small-business and startup clients — the same 0→1 breadth early-stage teams hire a generalist to provide.

Lifecycle ownership (concept → alpha → SEO/maintenance): - Sourced clients, ran discovery, translated vague desires into scoped concepts and diagrams (Balsamiq), drove design approval, built the alpha, shipped to production, and stayed for SEO, optimization, and ongoing maintenance. - Acted as designer, frontend, backend, DBA, deploy engineer, and support — directly analogous to employee #3–#10 wearing every hat.

What this proves for startups: - Comfort with ambiguity, direct customer contact, and accountability for uptime after launch — not just ticket execution. - Breadth from **server/workstation deployment (PC/Mac/Linux)** to **Arduino-based integrated controllers** to modern web apps — I debug across the whole stack. - Long-running client relationships where shipping fast and keeping it running matters more than stack purity.

Tailored for: Senior Software Embedded Engineer at Axon.

Lead Front End Developer

2FIX.US | Seattle, WA | June 2014 – Present (11+ years)

End-to-end web delivery for diverse small-business clients on modern React-based stacks.

- **Located clients and defined project goals/functionality** — requirements gathering from non-technical stakeholders, articulated trade-offs on style, scope, and timeline.
- **Designed initial concepts → built alpha → iterated on feedback → deployed production** — using Visual Studio Code, **React / JavaScript component architecture**, and supporting **SEO, maintenance, and content updates** post-launch.
- Headed production of JavaScript components; established patterns for reusable, maintainable frontend code that other developers could extend.
- Follow-up ownership: SEO optimization, performance tuning, CMS integration (WordPress/Drupal) where it delivered faster time-to-value — pragmatic tech selection over dogma.

Startup translation: I don't hand off after "dev done" — I own concept through live-in-prod and the long tail of iteration that determines retention.

AI-Forward Full-Stack Engineer (Independent)

AI Systems Portfolio — Production Builds | Seattle, WA | January 2024 – Present

Applied the same full-stack + ship-fast muscle to AI-augmented products. Framed as **product engineering** (not research): rapid prototyping in **days not months**, edge deployment, production hardening, and measurable cost/quality wins.

Delobotomize — Context Recovery System (Full-Stack + Edge)

Full-stack HNSW vector system that turns 2-hour fragmented sessions into 8+ hour deep work.

- Built **TypeScript + Bun + Redis + vector embeddings + REST APIs**, with **VS Code / terminal / browser integrations** for automatic capture; **sub-100ms semantic search** via HNSW indexing.
- Deployed capture/retrieval paths to edge where latency mattered; **recovery in <30s**, **98.7% success**, **10MB / 100K tokens** storage efficiency.
- **Impact:** 4× session extension (2h → 8h+), 87% less re-briefing (30m → 4m), **50K+ context saves/month**.
- Shipped with docs, video tutorials, and feedback loops — same training/adoption muscle startups need for new features.

DataKiln — Multi-Provider Workflow Automation (Hono / Cloudflare Workers / D1)

Cost-aware routing and orchestration framework deployed to 200+ edge locations.

- **Hono + Cloudflare Workers + D1 + TypeScript**; intelligent routing across **10+ AI providers** by cost/speed/capability, with **circuit breakers and adaptive rate limiting** for 98.5% success.
- **Edge deployment first-class:** global low-latency, zero-downtime blue-green deploys, edge caching where it cut cost.
- **Impact: $0.12 → $0.03 per operation (75% cut)**, **500+ ops/hour sustained**, **100K+ data points/day**, **99.9% uptime (6 months)**.
- Built cost/usage dashboard (API + monitoring) so non-technical operators could see where every dollar went — product thinking, not just plumbing.

Multi-Agent Research Orchestration (Full-Stack Automation)

Workflow engine turning 4-hour manual research into 45-minute automated pipelines (YouTube transcript → analysis → cited synthesis).

- Orchestrated specialized workers with **Claude Code CLI + Playwright browser automation + MCP servers + TypeScript**; integrated web extraction with structured synthesis and per-project cost tracking.
- **Impact: 80% time cut (4h → 45m)**, **300+ research documents** with **10–20 citations each**, **$0.03–$0.05/session**, **9.2/10 user satisfaction** — enabling non-technical researchers to self-serve.

Common thread: Prototype a working, *deployed* slice in days, instrument cost/latency, then harden — the pace early-stage teams live or die on.

Full-Stack Developer — Contract Work

Various Clients (E-commerce, Logistics, Data) | Remote | 2019 – 2023 (4 years)

Core full-stack evidence pulled forward for generalist roles — each project owned frontend + backend + data + deploy + client comms.

Representative shipped work: - E-commerce platform — Svelte / Node.js / PostgreSQL — 10K+ monthly users. Full-stack build: product catalog, cart/checkout, auth, admin, Postgres schema and query tuning, Node API, Svelte storefront, production deploy and monitoring. Proved ability to run a revenue-adjacent system end-to-end. - **Real-time logistics dashboard — React / WebSocket.** Live tracking UI with WebSocket push, React state management for high-frequency updates, backend aggregation service, and alerting. Built for operators who need truth *now*, not on refresh. - **Automated data pipelines — Python / AWS Lambda — 1M+ records/month.** Python extraction/normalization on Lambda, S3/DB sinks, retry/idempotency, and observability for unattended daily runs. Cost-conscious by design (Lambda + right-sized batching).

Skills exercised on every engagement: full-stack web dev (frontend + backend + DB), **API design and third-party integration**, production deployment and monitoring, and **client communication / requirements gathering** — scoping, estimating, and saying no to scope that doesn't serve the milestone.

EDUCATION

University of Washington — Coding Bootcamp | Seattle, WA | May 2019 – Aug 2019 *Intensive full-stack program — fast-paced, breadth-first by design. Technologies covered:* **HTML5, CSS3, SASS, JavaScript, jQuery, Bootstrap, Firebase, Node.js, MySQL, MongoDB/Mongoose, Express, Handlebars.js, React.js & React Native.** Built multiple deployed apps under deadline pressure — direct prep for startup pace.

University of Washington | Seattle, WA | 2002 **B.A., Comparative History of Ideas** — interdisciplinary research, critical analysis, and writing. Strong foundation for ambiguous problem framing and clear stakeholder communication.

TECHNICAL EXPERTISE — GROUPED FOR EARLY-STAGE TEAMS

Frontend — What the User Sees

Svelte, SvelteKit, React, React Native, TailwindCSS, HTML5/CSS3/SASS, responsive/accessible UI, component-driven architecture. Legacy breadth (jQuery, Bootstrap, Handlebars) useful when inheriting existing code.

Backend — What Keeps It Running

Hono, Express.js, Node.js, Bun, Python, REST APIs, WebSocket, MCP. API-first design; comfortable owning auth, rate limiting, queues, and background jobs.

Data — Where Truth Lives

PostgreSQL (schema design, query tuning), Redis (caching, sessions, rate limiting), D1 (Cloudflare edge), MySQL, MongoDB/Mongoose, Sequelize, Firebase, Vector DBs / HNSW embeddings.

Mobile

React Native — shared JS/TS codebase to iOS/Android; bridges to native APIs where needed.

Integration — The Startup Multiplier

Third-party API integration as a primary skill (Postman-driven discovery, OAuth, webhooks, pagination, retry/backoff, schema mapping). Browser automation (Playwright) for un-API'd workflows. CMS integration (WordPress/Drupal) when it ships faster than custom.

Deploy & Edge — Ship to Prod and Keep It Up

Production deployment (Vercel, Cloudflare Workers, AWS Lambda), **edge deployment** (200+ locations, edge caching, D1 at the edge), Docker, GitHub Actions CI/CD, blue-green deploys, monitoring, 99.9% uptime patterns, Linux/macOS/Windows administration.

AI-Forward — Pragmatic, Not Theoretical

Claude / GPT-4 / Gemini / local LLMs (Llama, Mistral), RAG, multi-agent orchestration, context management, token/cost optimization (60–80% reductions proven), evaluation harnesses. Value prop: *select the right model, integrate it into existing systems, train the team, measure ROI* — not building models from scratch.

Tools & Workflow

Git, GitHub Actions, Docker, Postman, VS Code, MySQL Workbench, Studio 3T, Balsamiq, Slack, Heroku, Zoom; Adobe Creative Cloud for asset work when designer bandwidth is thin.

WHY EARLY-STAGE TEAMS (2–50) CHOOSE ME

1. **Owner/Operator Breadth** — 17 years finding work, scoping it, shipping it, and supporting it in prod. I default to ownership, not tickets.
2. **Full Lifecycle in One Person** — concept → alpha → production → SEO → maintenance is my normal, not a stretch assignment.
3. **Rapid Prototyping in Days, Not Months** — working, *deployed* prototypes fast, then instrumented iteration on cost/latency/quality.
4. **Edge-Native Deployment** — Cloudflare Workers/D1 and global edge patterns are already in my shipped systems, not slideware.
5. **API Integration as a Superpower** — third-party APIs, OAuth, webhooks, and browser automation to stitch together whatever the business already uses.
6. **Production Hardening** — 99.9% uptime, circuit breakers, fallbacks, monitoring, and cost dashboards so "demo → prod" doesn't become an outage.
7. **Cost-Conscious** — 60–80% AI cost cuts, $50K+ annual savings, $0.12→$0.03/op — every integration includes cost tracking and right-sizing.
8. **Customer-Proximate** — years translating non-technical stakeholder desires into shippable scope and timelines; I run the meeting and ship the fix.

SELECTED BUILD LOG — SHIPPED, NOT SLIDES

Build	Stack	What Shipped	Signal
Delobotomize	TS, Bun, Redis, Vectors, REST	Context capture + sub-100ms retrieval, <30s recovery, 50K saves/mo	4× longer deep-work sessions
DataKiln	Hono, Workers, D1, TS	10-provider routing, circuit breakers, 200+ edge locations	75% cost cut, 500 ops/hr, 99.9% uptime
Research Orchestration	Claude CLI, Playwright, MCP	Transcript → analysis → cited doc pipeline	4h → 45m, 300+ docs, $0.03/session
E-commerce (Contract)	Svelte, Node, Postgres	Catalog + cart/checkout + admin, prod deploy	10K+ users/mo
Logistics Dashboard	React, WebSocket	Real-time tracking with live push	Operator-grade latency
Data Pipelines	Python, Lambda	1M+ records/mo automated ETL	Unattended, idempotent, observed

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OPEN SOURCE & WRITING

- **LM Studio — External Drive Model Storage Utility** (July 2024) — bash utility with symlink management for large model collections. Issue: `lmstudio-ai/lms#260`.
- **MCP Server Examples** — example servers demonstrating integration best practices (in progress).
- **300+ technical documents** on AI implementation patterns and system architecture; **5–15 research syntheses/day** during portfolio builds.
- **Planned:** "Cost Optimization Strategies for Production LLM Deployments," "Multi-Agent Systems: Practical Implementation Patterns."

LANGUAGES & ADDITIONAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, SQL, Bash, PHP; fluent in **Spanish**
- **Integrations:** Third-party APIs, OAuth, CMS (WordPress/Drupal)
- **Hardware/Systems:** PC/Mac/Linux server & workstation deployment, Arduino-based controllers — useful when the startup's "full-stack" includes the physical world
- **Methods:** Agile methodologies, rapid prototyping, stakeholder communication, cost/ROI modeling

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AVAILABILITY & PREFERENCES

Applying for: Senior Software Embedded Engineer at Axon - **Full-Stack Engineer / Startup Generalist / Software Engineer (Early Stage)** — full-time, Seattle-area - Early-stage teams **(2–50 employees)** where one engineer covers frontend, backend, data, deploy, and customer lifecycle - Hybrid / on-site in Seattle or remote for Seattle-anchored teams; comfortable with in-person customer discovery

Ideal Team: - Small, senior-leaning, shipping-focused — values breadth, ownership, and "prototype in days" pace over narrow specialization - Has real customers and real production systems (or is about to) — needs someone who can own concept → prod → maintenance - Sees AI as a **feature accelerator** (integrate the right model via API, measure cost/quality, ship), not a research lab

Not Optimizing For: - Large-enterprise ticket queues with narrow ownership - Pure AI research (training models from scratch) - Roles without production deployment and customer proximity

Location: Seattle, WA — 1710 NW 57th St. Apt. 501, Seattle, WA 98107 — (206) 409-2846

CONTACT

📧 **Email:** aaaronmiller@gmail.com 💼 **LinkedIn:** linkedin.com/in/aaaronmiller 🔗 **GitHub:** github.com/aaaronmiller 🌐 **Portfolio:** aaaronmiller.github.io/portfolio

Response time: Within 24 hours

APPENDIX: METRICS THAT MATTER TO STARTUPS

Shipping Speed

- Working, deployed AI-forward slices in **days, not months** (Delobotomize, DataKiln, Research Orchestration each built iteratively in weeks with usable demos in days)
- Client lifecycle at 2FIX.US: concept → approved design → alpha → feedback iteration → prod + SEO — run repeatedly for 11+ years

Reliability & Scale Already Proven

- **99.9% uptime** across production AI implementations (6-month window); 98.5% success at 500+ ops/hour
- **300M+ tokens** processed; **100K+ data points/day**; **1M+ records/month** (Python/Lambda pipelines)
- **Sub-100ms retrieval (p95)**, **800ms p50 / 2.1s p95** operation latency, **10MB / 100K tokens** storage efficiency

Cost Discipline

- **60–80% AI cost reduction** via platform selection and workflow optimization; **$50K+ annual savings** for clients; **$0.12 → $0.03/op**
- Per-session and per-project cost tracking built into every system — no surprise bills

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Forked from AI Implementation Specialist base CV (2025-12-11) and reframed for startup generalist roles. All systems and metrics are shipped and measurable; demos and references available on request.